

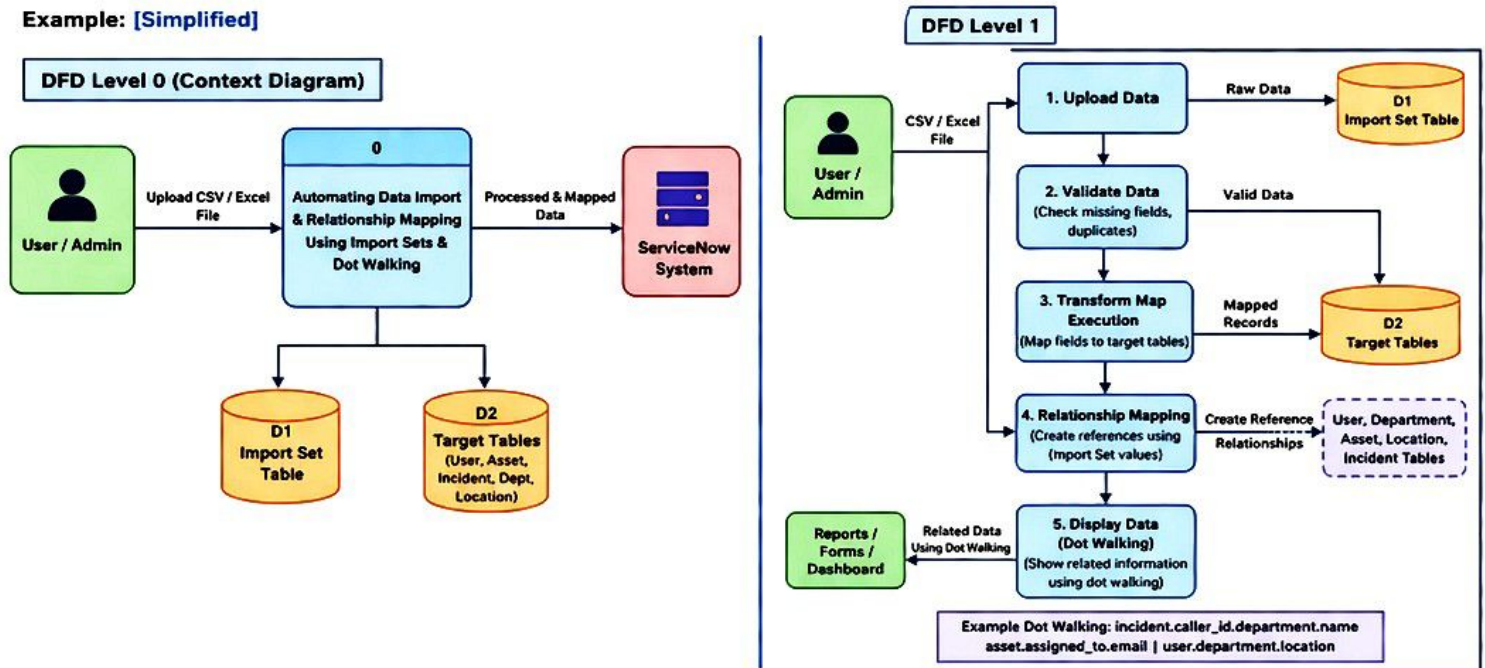
Project Design Phase-II Data Flow Diagram & User Stories

Date	31 January 2025
Team ID	SWTID-2026-3651
Project Name	Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [Simplified]



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	I can access my account / dashboard after successful registration.	High	Sprint-1
		USN-2	As a user, I will receive confirmation email once I have registered for the application.	I can receive confirmation email & click the confirm link.	High	Sprint-1
		USN-3	As a user, I can register for the application through Facebook.	I can register & access the dashboard with Facebook Login.	Low	Sprint-2
		USN-4	As a user, I can register for the application through Gmail.	I can register & access the dashboard with Google account.	Medium	Sprint-1
	Login	USN-5	As a user, I can log into the application by entering email & password.	I am logged in and redirected to the dashboard.	High	Sprint-1
	Dashboard	USN-6	As a user, I can view dashboard with my import summary, related records and quick actions.	Dashboard shows correct counts of Users, Assets & Incidents with relationship details using dot walking.	Medium	Sprint-2
Customer (Web user)	View Imported Data	USN-7	As a web user, I can search and view imported records with related information using dot walking.	I can see related details like incident caller's department or asset owner's email without extra coding.	Medium	Sprint-2
Customer Care Executive	Validate & Correct Data	USN-8	As a support executive, I can identify and correct failed or invalid import records.	Invalid records are listed with error messages and can be corrected & re-imported successfully.	High	Sprint-2
Administrator	Manage Import Sets & Transform Maps	USN-9	As an administrator, I can create, schedule and run import sets & transform maps to automate bulk data import.	Import runs successfully on schedule and data is inserted/updated in target tables with proper relationships.	High	Sprint-1

Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

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Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	- Registration through Form - Registration through Gmail - Registration through LinkedIn
FR-2	User Confirmation	- Confirmation via Email - Confirmation via OTP
FR-3	Profile Management	- Edit user profile - View user profile
FR-4	Data Import & Mapping	- Import data from Excel files - Map data fields with system fields - Automated data import and relationship mapping using dot walking

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Intuitive and easy-to-use interface.
NFR-2	Security	Data encryption, secure authentication, and access controls.
NFR-3	Reliability	System must function correctly without failure.
NFR-4	Performance	Fast response times and quick processing of data.
NFR-5	Availability	System should be available 24/7 with minimal downtime.
NFR-6	Scalability	Ability to handle increasing numbers of users and data.

Project Design Phase-II Technology Stack (Architecture & Stack)

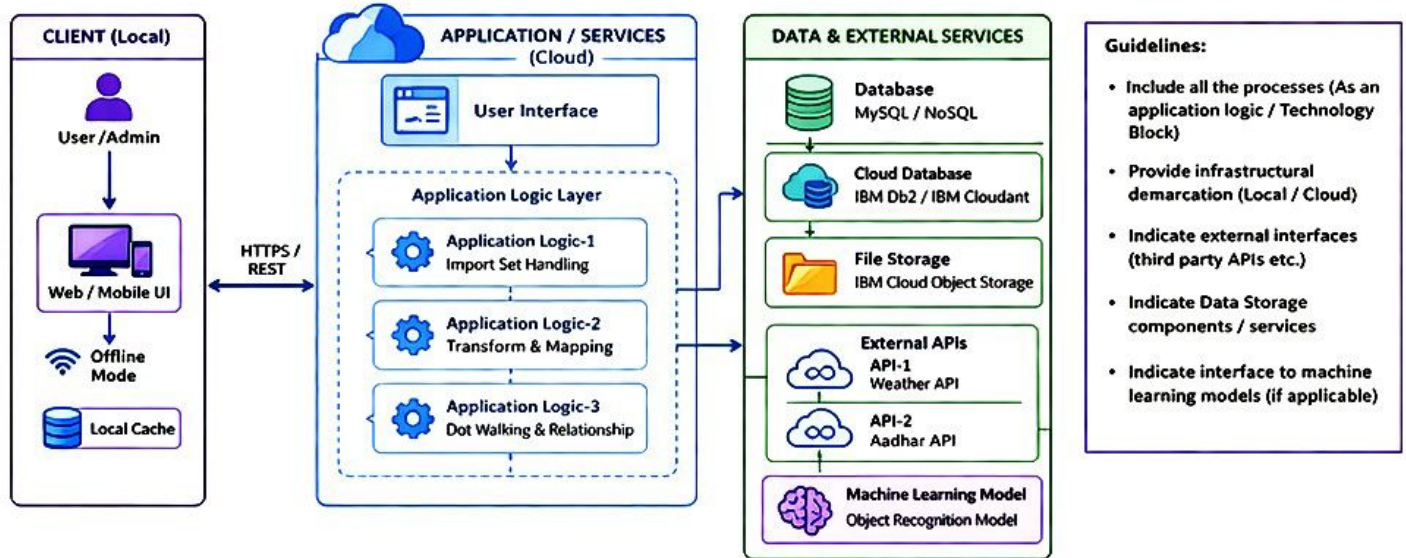
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Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2.

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



S.No	Component	Description	Technology
1.	User Interface	How user interacts with application (Web, Mobile, Chatbot etc.)	HTML, CSS, JavaScript / Angular.js / React.js
2.	Application Logic-1	Import Set creation, validation and loading	Java / Python (ServiceNow)
3.	Application Logic-2	Transform Map execution and field mapping	IBM Watson STT Service / JavaScript
4.	Application Logic-3	Relationship creation and dot walking for display	IBM Watson Assistant / JavaScript
5.	Database	Stores application master and transaction data	MySQL, NoSQL
6.	Cloud Database	Managed database service on cloud	IBM Db2, IBM Cloudant
7.	File Storage	Store uploaded files and import set files	IBM Cloud Object Storage / Local File System
8.	External API-1	Weather information retrieval	IBM Weather API
9.	External API-2	Aadhar verification service	Aadhar API
10.	Machine Learning Model	Image / document recognition support	Object Recognition Model (Custom)
11.	Infrastructure (Server / Cloud)	Application deployed on cloud platform. Local Server: ServiceNow Instance (Personal Dev) Cloud: ServiceNow Cloud / IBM Cloud	Local, Cloud Foundry, Kubernetes

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source components used for development	ServiceNow Platform, Node.js, Angular.js, Bootstrap
2.	Security Implementations	Authentication, authorization, encryption, and access controls	SSL/TLS, SHA-256, Role Based Access Control (RBAC), ServiceNow ACL
3.	Scalable Architecture	Architecture supports horizontal scalability (3-tier / Microservices)	ServiceNow Cloud, REST APIs, Load Balancer
4.	Availability	Application should be highly available with minimal downtime	ServiceNow High Availability, Cloud Redundancy, Backup & Failover

S.No	Characteristics	Description	Technology
5.	Performance	Designed to handle bulk imports and mapping with optimized transform maps, caching, and indexing	Transform Map Optimization, DB Indexing, Caching, CDN (Cloudflare)